MINI PROJECT 1
BUILD YOUR OWN CYBERSECURITY LAB

Student Details	Information
Name	Neralla Sairam Suraj
Student Code	STU-B33F7C
Host OS	Windows 10
Virtualization Software	VMware Workstation
Guest OS	Kali Linux (Kali GNU/Linux Rolling, VERSION_ID 2024.3 as shown in terminal)

1. Aim

To set up and document a cybersecurity practice environment using Windows 10 as the host operating system, VMware Workstation as the virtualization platform, and Kali Linux as the guest operating system.

2. Objectives

- Configure and run a Kali Linux virtual machine in VMware Workstation on Windows 10.
- Check the guest OS version, architecture, memory and network interface using Linux commands.
- Record the VM hardware configuration and network mode.
- Use a virtual lab responsibly for introductory cybersecurity learning.

3. Tools Used

- Windows 10 — host operating system.
- VMware Workstation — creates and runs the guest VM.
- Kali Linux — guest operating system for security learning.
- Kali terminal — used for OS and network verification.

4. System Configuration

Component	Verified detail
Host OS	Windows 10 (as specified by student)
Virtualization	VMware Workstation; VM hardware compatibility shows Workstation 17.5.x
VM name	Kali 2026 x64 Customized by zSecurity v1.0.0
Guest OS	Kali GNU/Linux Rolling; VERSION_ID 2024.3 from terminal output
Architecture	x86_64 (uname -m)
VM memory	2 GB configured; free -h reports 1.9 GiB visible to guest
Virtual disk	80 GB (VMware summary)
Processors	2 (VMware summary)
Network mode	NAT (VMware summary)

5. VMware Workstation Configuration Evidence

The screenshot below was supplied from the running VMware Workstation environment. It shows the VM name, assigned resources and NAT network adapter.

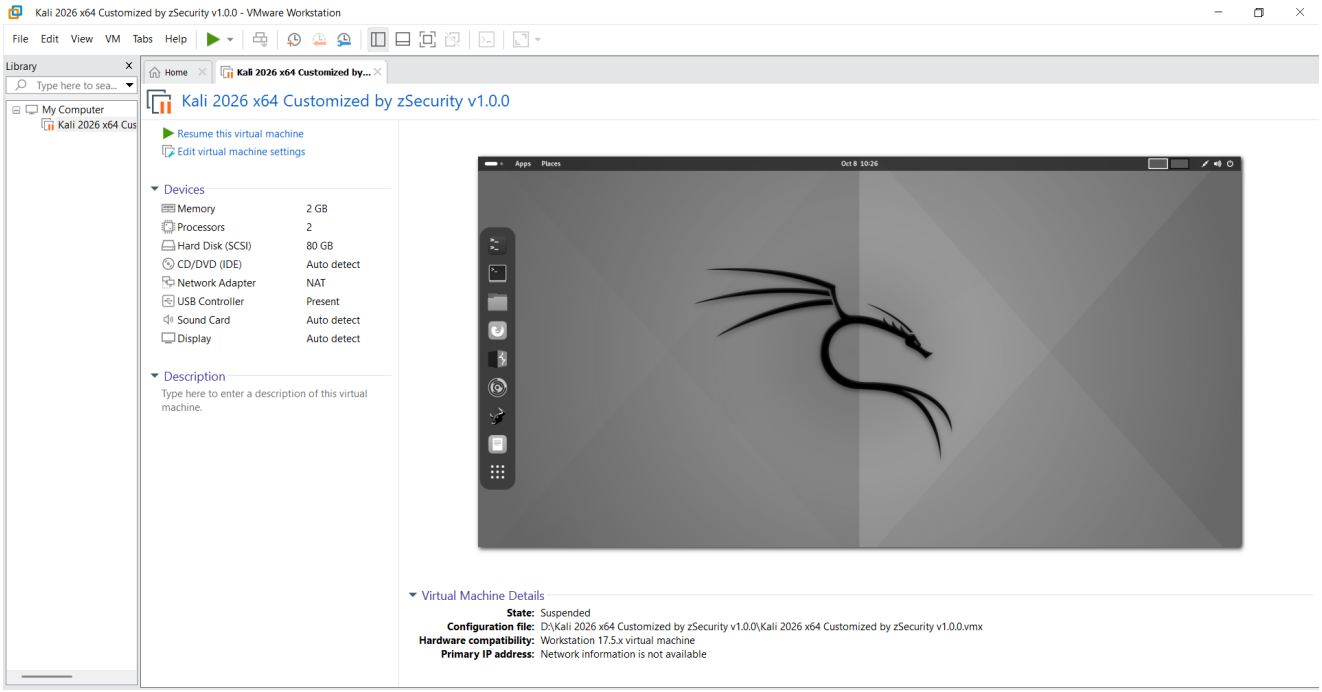


Figure 1. VMware Workstation summary for the Kali Linux virtual machine (actual screenshot supplied by student).

6. Kali Linux Desktop Evidence

The following screenshot shows the Kali Linux desktop running inside the VMware Workstation window.

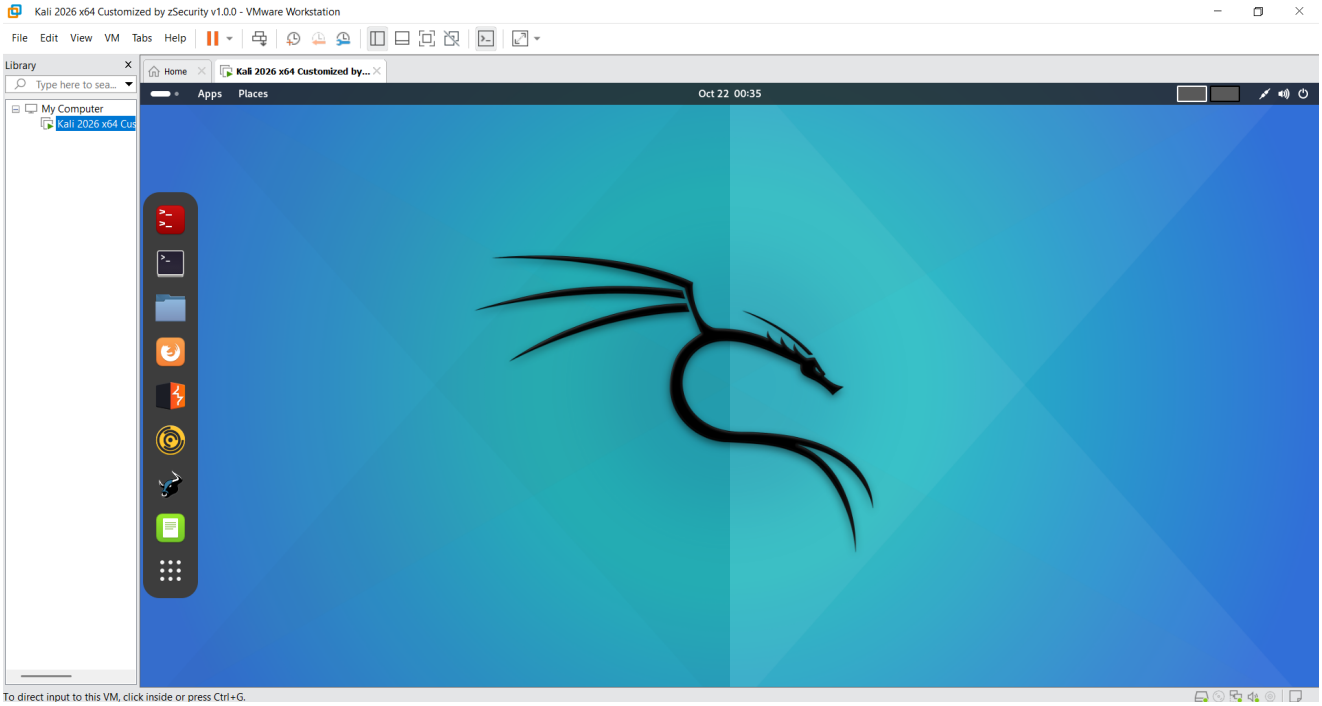


Figure 2. Kali Linux desktop running as a guest in VMware Workstation (actual screenshot supplied by student).

7. Installation and System Verification

The student ran the following commands in the Kali Linux terminal to verify the guest environment:

Command	Purpose	Result shown in screenshot
cat /etc/os-release	Identify the Linux distribution and release	Kali GNU/Linux Rolling; VERSION_ID=2024.3
uname -m	Check kernel architecture	x86_64
free -h	Check memory visible to the guest	1.9 GiB total; 1.1 GiB available at capture time
ip addr	Inspect network interfaces	eth0 is UP; IPv4 address 192.168.64.128/24

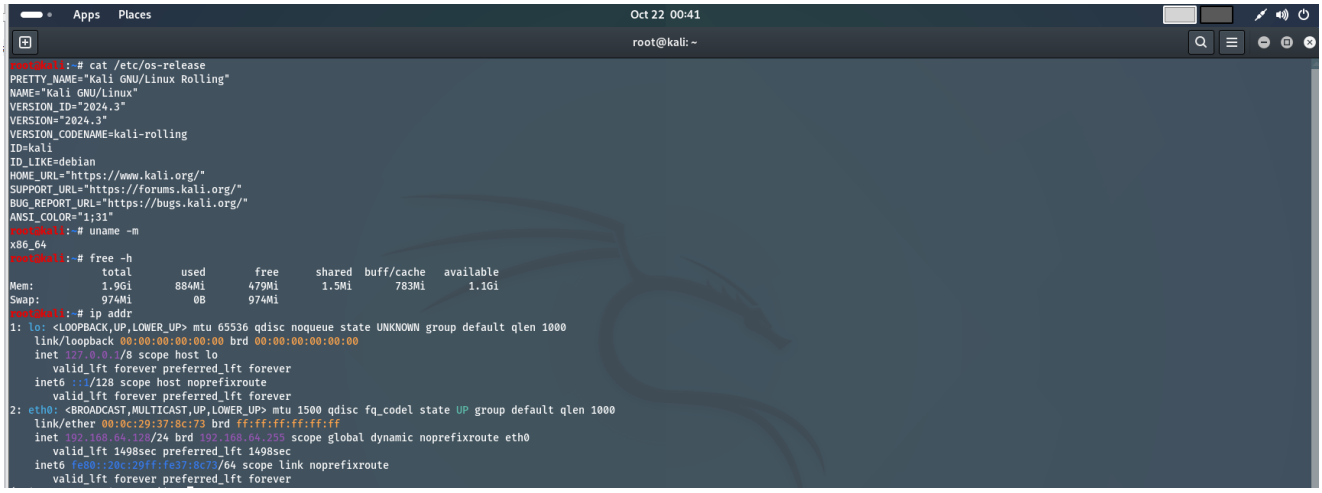


Figure 3. Actual Kali terminal output showing OS release, architecture, memory and network interface

8. Procedure

- Open VMware Workstation on the Windows 10 host.
- Select the configured Kali Linux virtual machine and resume/start it.
- Sign in to the guest and open a terminal.
- Run each verification command separately and inspect the output.
- Capture screenshots of the VM summary, guest desktop and terminal output.

9. Network Configuration and Safety

The VM summary shows NAT networking. NAT generally allows the guest to access external networks through the host's connection while using a virtual network interface. The actual guest interface shown in the terminal is eth0.

- Use NAT for general lab tasks that need internet connectivity; use host-only networking when a lab should remain isolated.
- Avoid bridged networking for vulnerable lab machines unless the exercise requires it and the network is controlled.
- Only test systems for which you have explicit permission.
- Do not include passwords, tokens or private credentials in report screenshots.
- Virtualization reduces risk but does not guarantee complete isolation; be cautious with shared folders, clipboard sharing and USB passthrough.

10. Observations

Check	Observation
VM startup	Kali Linux desktop is visible in VMware Workstation.
Operating system	Kali GNU/Linux Rolling, version 2024.3 as reported by /etc/os-release.
Architecture	x86_64.
Allocated resources	2 processors, 2 GB memory and 80 GB virtual disk according to VMware summary.
Memory check	free -h reports 1.9 GiB total memory visible to the guest at capture time.
Network	NAT selected; eth0 is UP with IPv4 address 192.168.64.128/24 in the captured output.

11. Result

A Kali Linux virtual machine was started in VMware Workstation, and the supplied screenshots verify its desktop, configured VM resources, OS release, x86_64 architecture, memory report and network interface. The recorded results correspond to the screenshots provided for this report.

12. Conclusion

This project documents a Windows 10 host running Kali Linux as a guest in VMware Workstation. The actual VM summary and terminal evidence were used to record the configuration and verify the guest environment. The setup provides a virtual environment for learning Linux administration and authorized cybersecurity practices.

13. References

- VMware Desktop Hypervisor: https://www.vmware.com/products/desktop-hypervisor/workstation-and-fusion
- Kali Linux downloads: https://www.kali.org/get-kali/
- Kali Linux documentation: https://www.kali.org/docs/
- Linux man-pages: https://man7.org/linux/man-pages/